CS & IT ENGINEERING



Algorithms

Divide & Conquer



Lecture No.- 07

By- Aditya sir

Recap of Previous Lecture



Topic

Topic

QS + MS → Questions

Matrix Multi

Topics to be Covered



- Topic
- Topic
- Topic

Matrix Multi

Masters Mtd



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW



Topic : Divide & Conquer

Matrix Multiplication Problem

w.r.t. square matrix of size $n \times n$.



Topic : Divide & Conquer

Basics : $A_{n\times n} * B_{n\times n} \rightarrow C_{n\times n}$

In general : $A_{r1\times c1} * B_{r2\times c2} = C_{r1\times c2}$

A and B are multiplicable only if $\boxed{c_1 = r_2}$



Topic : Divide & Conquer

Eq : Matrix Addition

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2\times2} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}_{2\times2} = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}_{2\times2}$$

$$A_{n\times n} + B_{n\times n} \rightarrow C_{n\times n}$$



Topic : Divide & Conquer

Time Complexity of Previous Method of Matrix Addition

$A_{n\times n} + B_{n\times n} \Rightarrow$

```
for i : 1 → n
    for j : 1 → n
        C[i, j] = A[i, j] + B[i, j]
```

$TC \Rightarrow \boxed{O(n^2)}$ [Matrix Addition]

and

Matrix subtraction



Topic : Divide & Conquer

Matrix Multiplication :

Given two square matrices each → $n \times n$

$$\underset{A}{\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}} * \underset{B}{\begin{bmatrix}5 & 6\\ 7 & 8\end{bmatrix}} = \begin{bmatrix}\underset{C_{11}}{1\times5+2\times7} & \underset{C_{12}}{1\times6+2\times8}\\ \underset{C_{21}}{3\times5+4\times7} & \underset{C_{22}}{3\times6+4\times8}\end{bmatrix}_{2\times2}$$



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Non D&C

Matrix multiplication in general : (~~Non D and C~~)

$$\underset{A}{\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}} * \underset{B}{\begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}} = \underset{C}{\begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}}$$

Then, $t_1 \rightarrow$

		Add	mul
$c_{11} = a_{11} * b_{11} + a_{12} * b_{21}$	$\rightarrow$	1	2
$c_{12} = a_{11} \times b_{12} + a_{12} \times b_{22}$	$\rightarrow$	1,	2
$c_{21} = a_{21} \times b_{11} + a_{22} \times b_{21}$	$\rightarrow$	1,	2
$c_{22} = a_{21} \times b_{12} + a_{22} \times b_{22}$	$\rightarrow$	1,	2

4 Addition & 8 Multi.



Topic : Divide & Conquer

Code of prev matrix multiplication logic (Non DnC)

$A_{n\times n} + B_{n\times n} = C_{n\times n}$

```
For i : 1 → n :
    For j : 1 → n :
        For k : 1 → n
            C[i, j] += A[i, k] * B[k, j]
```

⇒ O(1) space

Can ~~are~~ do better using DnC approach >

⇒ $TC = O(n^3)$



Topic : Divide & Conquer

Eg : Divide & Conquest based approach

16 × 16

8	8
8	8

A_{11} A A_{12}

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 7 \\ \hline 9 & 2 & 1 & 3 \\ 5 & 7 & 2 & 4 \end{bmatrix}_{4\times4} * \begin{bmatrix} 3 & 4 & 5 & 6 \\ 7 & 8 & 9 & 2 \\ \hline 3 & 1 & 7 & 4 \\ 3 & 2 & 9 & 8 \end{bmatrix}_{4\times4} \Rightarrow \begin{bmatrix} C_{11} & C_{12} \\ \hline C_{21} & C_{22} \end{bmatrix}$$

A_{21} A_{22} B_{11} B B_{12} C B_{21} B_{22}

$n \times n$

$\downarrow$

$n/2 \times n/2$

$\Rightarrow T(n)$

$\Downarrow$

8 mat mul

4 mat additn

~~16 × 16 → 4 × 4~~

~~4 × 4 → 2 × 2~~

Small problem



Topic : Divide & Conquer

Sub-problem

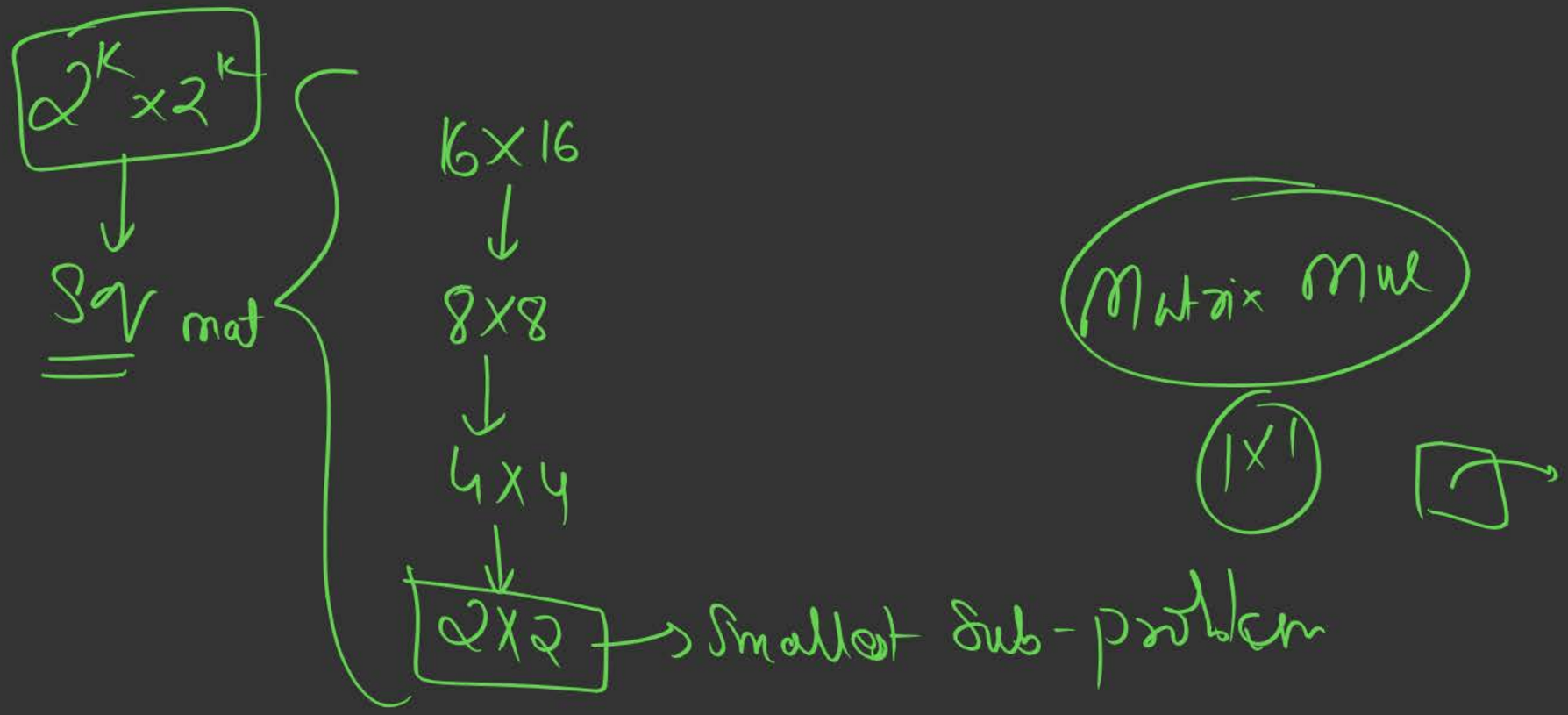
Where,

	Add	Mul
$C_{11} = A_{11} * B_{11} + A_{12} * B_{21} - (1)$	1,	2
$C_{12} = A_{11} * B_{12} + A_{12} * B_{22} - (2)$	1,	2
$C_{21} = A_{21} * B_{11} + A_{22} * B_{21} - (3)$	1,	2
$C_{22} = A_{21} * B_{12} + A_{22} \times B_{22} - (4)$	1,	2

4 sub-matrix addition
&
8 sub-matrix Mul

$A, B \rightarrow n \times n$

Subproblem $\rightarrow$ Divide


$2^k \times 2^k$
Sq mat
16×16
8×8
4×4
2×2
smallest sub-problem
Matrix mul
1×1



Topic : Divide & Conquer

$T(n) = 2T(n/2) + O(n)$

Divide :

$n \times n \rightarrow \left(\frac{n}{2} \times \frac{n}{2}\right)$ Subproblem

↓

Original problem

↓

$n \times n$

↓

$\frac{n}{2} \times \frac{n}{2}$

D&C
DP

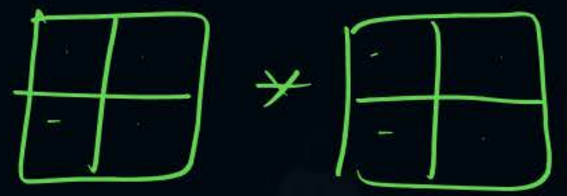
TC RC
TC

$T(n) = 8 * T(n/2) + O(n^2)$ → gate



Topic : Divide & Conquer

Let T(n) represent the time complexity to multiply two square matrices A & B each of size $n \times n$. (square matrix) → Recurrance for DnC based matrix multiplication.



* $T(n) = C, \quad n \leq 2$

$T(n) = 8\,T(n/2) + b * n^2, \quad n > 2, \quad b > 0$

$T(n/2) = 8T\,(n/2^2) + b * n^2/4$

$$T(n) = 8\left[8T\left(\frac{n}{2^2}\right) + \frac{bn^2}{4}\right] + b * n^2$$

$$= 8^2T\left(\frac{n}{2^2}\right) + 2bn^2 + bn^2$$

$$= 8^2T\left(\frac{n}{2^2}\right) + 3bn^2 \qquad \ldots. (2)$$

$2^k = n$

$(2^k)^3 = n^3$

$2^{3k} = n^3$

$(2^3)^k = n^3$

$$T(n) = 8^2 T(n/2^2) + 3bn^2$$

$$= 8^2\left[8T(n/2^3) + b \times \left(\frac{n}{2^2}\right)^2\right] + 3bn^2$$

$$= 8^3 T(n/2^3) + \frac{8^2 \times bn^2}{2^4} + 3bn^2$$

$$= 8^3 T(n/2^3) + \frac{64}{16} bn^2 + 3bn^2$$

$$= 8^3 T(n/2^3) + 4bn^2 + 3bn^2$$

$$\boxed{= 8^3 T(n/2^3) + 7bn^2}$$

$2^k = n$

8^k

$\Rightarrow (2^3)^k = (2^k)^3$

$= n^3$



Topic : Divide & Conquer

General Term :

$$T(n) = 8^k T\left(\frac{n}{2^k}\right) + (2^k - 1) * bn^2 \quad \ldots (2)$$

Let $\frac{n}{2^k} = 1 \Rightarrow 2^k = n \Rightarrow k = \log_2 n$

$2^k \rightarrow n$

$8^k \rightarrow n^3$

$$T(n) = n^3 T(1) + (n-1) * bn^2$$

$$= n^3 * c + (n-1) * bn^2$$

$$T(n) = n^3 * c + bn^3 - bn^2 \Rightarrow O(n^3)$$

TC after using this simple DnC based approach is also $O(n^3)$.

$8^k = n^3$

$\frac{n}{2^k} = 2$

$n = 2^k * 2$

$n = 2^{k+1}$

$k+1 = \log_2 n$

$k = \log_2 n - 1$



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Observations :

1. In DnC based matrix multiplication, there are 8 subproblems (sub-matrix multiplication), involved in getting C_{11}, C_{12}, C_{21}, C_{22}.
2. Time complexing will only be reduced, if the no. of sub-matrix multiplications are reduced from 8 to a smaller value.

$T(n) = 8T(n/2) + O(n^2)$

$(* \rightarrow +)$

⇓

"STRASSEN" → Research

Logic :

$a * b = \underbrace{a + a + \ldots\ldots + a}_{b \text{ times}}$

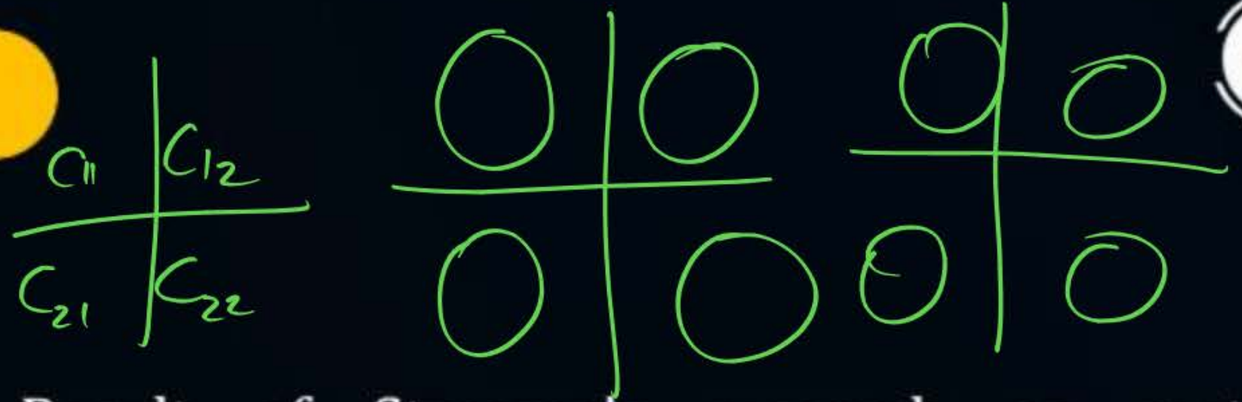


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Stras Sen's Matrix Multiplication

$$P = (A_{11} + A_{22}) \cdot (B_{11} + B_{22})$$
$$Q = (A_{21} + A_{22}) \cdot B_{11}$$
$$R = A_{11} \cdot (B_{12} - B_{22})$$
$$S = A_{22} \cdot (B_{21} - B_{11})$$
$$T = (A_{11} + A_{12}) \cdot B_{22}$$
$$U = (A_{21} - A_{11}) \cdot (B_{11} + B_{12})$$
$$V = (A_{12} - A_{22}) \cdot (B_{21} + B_{22})$$

$$C_{11} = P + S - T + V$$
$$C_{12} = R + T$$
$$C_{21} = Q + S$$
$$C_{22} = P + R - Q + U$$



Result of Strassen's research on matrix multiplications.

$\{A_{n\times n} * B_{n\times n} \rightarrow C_{n\times n}\}$ → Original problem

$$A_{ij}, B_{ij}, C_{ij} \Rightarrow \frac{n}{2} \times \frac{n}{2}$$

Additionally created/used sub-matrices

$$P, Q, R, S, T, V, U \Rightarrow \left(\frac{n}{2} \times \frac{n}{2}\right)$$



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Let T(n) represents the TC for the matrix multiplication using Strassen's approach :

$$T(n) = c, \quad n \leq 2$$

$$T(n) = 7 \times T\left(\frac{n}{2}\right) + bn^2, n > 2$$

Strassen's Recurrence

$$T(n/2) = 7 * T\left(\frac{n}{2^2}\right) + b\left(\frac{n}{2}\right)^2$$

$$T(n) = 7\left[7T\left(\frac{n}{2^2}\right) + \frac{bn^2}{2}\right] + bn^2$$

$$= 7^2T\left(\frac{n}{2^2}\right) + \frac{7bn^2}{4} + bn^2 * (7/4)^0$$



Topic : Divide & Conquer

General Term :

$$T(n) = 7^k T\left(\frac{n}{2^k}\right) + bn^2 * \left(\sum_{i=0}^{k-1} \left(\frac{7}{4}\right)^i\right)$$

GP : $\sum_{i=1}^{n} x^i < x^{n+1}$

$$T(n) = 7^k T\left(\frac{n}{2^k}\right) + bn^2 * \left(\frac{7}{4}\right)^k$$

$$= 7^k T(1) + bn^2 * \left(\frac{7}{4}\right)^k$$

$$= 7^k * c + bn^2 * \frac{7^k}{4^k}$$

$2^k = n$

$4^k = n^2$

$$\sum_{i=0}^{3} = 2^1 + 2^2 + 2^3 + 2^0$$

$= 2 + 4 + 8 + 1$

$= 15 < 2^k$

$= 15 < 16$

$n/2^k = 1$

$2^k = n$

$k = \log_2 n$



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$T(n) = 7^k * c + bn^2 * \frac{7^k}{n^2}$

$2^k = n$

$4^k = n^2$

$T(n) = 7^k * c + 7^k * b$

$= 7^k (b + c)$

$T(n) = d * 7^k$

Let $b + c = d$ (constant)

$k = \log_2 n$

$7^k = 7^{(\log_2 n)}$

$= n^{\log_2 7}$

$= n^{2.81}$

$\log_2 7 = 2.81$

$7^{\log_2 n}$

$= n^{(\log_2 7)}$

$T(n) = O(7^k)$

$T(n) = O(n^{2.81})$

$n^3 \Rightarrow n^{2.81}$

$(103)^3 > (103)^{2.81}$

0.19



Topic : Divide & Conquer

Complexity Analysis of matrix mul (square) : Time complexity summary

1. Non – DnC ⇒ $O(n^3)$
2. Simple DnC ⇒ $O(n^3)$
3. Strassen's based DnC ⇒ $O(n^{2.81})$

$T(n) = 8T(n/2) + O(n^2)$

$T(n) = 7T(n/2) + O(n^2)$

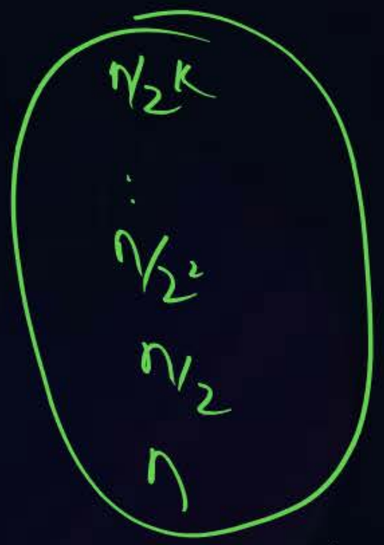


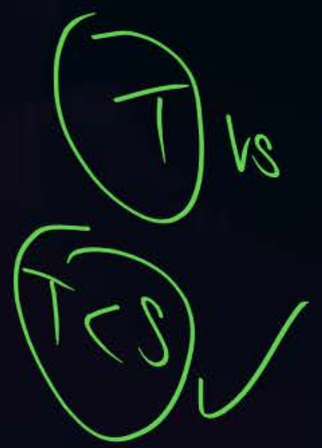
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Space Complexity :

1. Hon – DnC ⇒ O(1) → Constant
2. Simple DnC ⇒ Recurrsion stock → $O(\log_2 n)$
3. Strassen's DnC ⇒ $O(\log n + n^2) = O(n^2)$ → P, Q, R, S, R, U, V

$O(n^3) \Rightarrow O(n^{2.81})$





(6) Master's Method:



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Solving DnC Recurrence

1. Back Substitution method → Universal method
   - (i) Value of recurrence
   - (ii) TC of recurrence → O( )
   - Min-max prob :
     - Value : $\left[\frac{3n}{2} - 2\right]$
     - TC → O(n)
2. Master's method/Theorem → Shortcut
   - (i) Only gives TC but not value
   - (ii) Works only in specific cases → (3 cases) → Memorization
3. Recurrence tree approach :
   - (i) Asymmetric DnC recurrence



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Master's method : Only applicable for

$$T(n) = a * T(n/b) + F(n)$$

Such that $a \geq 1, b > 1, F(n) = +ve$ → Symmetric DnC

Special case (later)

Case 1 : If $F(n)$ is $O(n^{(\log_b a - E)})$, for same $E > 0$,

Then $T(n) = \theta(n^{\log_b a})$

$T(n) = 8T(n/2) + n^2$

$T(n) = aT(n/b) + F(n)$

$a = 8$

$b = 2$

$f(n) = n^2$



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Case 2 : If $F(n)$ is $\theta\left(n^{\log_b a} * (\log n)^K\right)$ for some K, such that

(a) $K \geq 0$, then $T(n) = \theta\left(n^{\log_b a} * (\log n)^{K+1}\right)$

(b) $K = -1$, then $T(n) = \theta\left(n^{\log_b a} * \log(\log n)\right)$



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$\log_b a = y$

Case 3 : If $F(n)$ is $\Omega(n^{(\log_b a + E)})$

For some $E > 0$

AND

$a * F(n/b) \leq \delta * f(n)$, For some $\delta < 1$

Then,

$T(n) = \theta(F(n))$

Case 1 : If $F(n) = O($

$T(n) = aT(n/b) + f(n)$

$a \geq 1,\ b > 1,\ f(n) = +ve$

✓ Case 1 :- $f(n) = O(n^{\log_b a - \varepsilon})$, $\varepsilon > 0 \implies T(n) = \Theta(n^{\log_b a})$

✓ Case 2 :- $f(n) = \Theta(n^{\log_b a} * (\log n)^k)$, for some k:

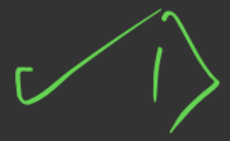
a) $k \geq 0$, $T(n) = \Theta(n^{\log_b a} * (\log n)^{k+1})$

b) $k = -1$, $T(n) = \Theta(n^{\log_b a} * \log(\log n))$

✓ Case 3 :- $f(n) = \Omega(n^{\log_b a + \varepsilon})$, some $\varepsilon > 0$

and $a * f(n/b) \leq \delta * f(n)$, some $\delta < 1$

$\Rightarrow T(n) = \Theta(f(n))$



$$T(n) = 4T(n/2) + n$$



Topic : Divide & Conquer

#Q. $T(n) = 4T(n/2) + n$

Check master's method form :

$$T(n) = a * T(n/b) + F(n)$$

$a = 4, b = 2, F(n) = n \quad \Rightarrow \quad$ Valid

$\log_b a = \log_2 4 = \log_2 2^2 = 2$



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Check Case 1 : If $F(n) = O(n^{2-E})$, some $E > 0$

Is $n = O(n^{2-E})$, $E > 0$

$E = 0.5, 0.1, 0.2, ...$

$n = O(n^{1.5})$ True

$n = O(n^{2-\varepsilon})$

Case 1 satisfied

Hence $T(n) = \theta(n^2)$

H.W : Recheck using Back - sub



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Check Using Back-substitution :

$T(n) = 4T(n/2) + n$

$T(n/2) = 4T(n/2^2) + n/2$

$T(n) = 4(4T(n/n^2) + n/2) + n$

$= 4^2\, T(n/2^2) + 2n + n$

$= 4^2\, T(n/2^2) + 3n$

$= 4^3\, T(n/2^3) + 7n$

.

.

.

$T(n) = 4^K T(n/2^K) + (2^K - 1)n$

$n/2^K = 1$

$2^K = n \quad \Rightarrow \quad 4^K = n^2$

$T(n) = n^2 T(1) + (n - 1)n$

$= n^2 * c + n^2 - n$

$T(n) = \theta(n^2)$

$$T(n) = 2T(n/2) + n \log n.$$



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#Q. $T(n) = 2T\,(n/2) + n \log n$

Check : $a = 2, b = 2, F(n) = n \log n$ $\Rightarrow$ Valid

$\log_b a = \log_2 2 = 1$

Check case 1 :

If $F(n) = O\left(n^{\log_b a - E}\right)$, $E > 0$

$n\log n = O\left(n^{1-E}\right)$, some $E > 0$?

Fails



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Check Case 2 :

If $F(n) = \theta\left(n^{\log_b a} * (\log n)^K\right)$

Case 2 : a) $K \geq 0$

$n\log n = \theta\left(n^1 * (\log n)^K\right)$ → True for $K = 1$

Case (a) satisfied.

hence, $T(n) = \theta\left(n^1 * (\log n)^{K+1}\right)$, $K = 1$

$= \theta(n * (\log n)^{1+1})$

$T(n) = \theta(n * (\log n)^2)$



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#Q. $T(n) = T(n/3) + n$

A $T(n) = \theta(n \log n)$

B $T(n) = \theta(n)$

C $T(n) = \theta(n^2)$

D $T(n) = \theta(\log n)$



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$T(n) = T(n/3) + n$

Validity :

$a = 1, b = 3, F(n) = n$

$\log_b a = \log_3 1 = 0$



Topic : Divide & Conquer

Case 1 :

If $F(n) = O\left(n^{\log_b a - E}\right)$, $E > 0$

$n = O(n^{0-E})$, some $E > 0$

Fails

Case 2 :

$n = \theta\left(n^{o} * (\log n)^{K}\right)$, some K

(a) $K \geq 0$ → Fails

(b) $K = -1$ → Fails



Topic : Divide & Conquer

$f(n) = n$
$P(n/3) = n/3$

Case 3 :

If $F(n) = \Omega\left(n^{\log_b a + E}\right)$, $E > 0$

$n = \Omega\left(n^{0+E}\right)$, $E > 0$?

$n = \Omega(n^{0.1})$ ✓ $E = 0.1, 0.5, 0.2, \ldots\ldots\ n \geq n^E, E > 0$

and

$a * F(n/b) \leq \delta * F(n)$, some $\delta < 1$

$1 * n/3 \leq 1/3 * n$, some $\delta < 1$

let take $\delta = 1/3$

$1 * n/3 \leq 1/3 * n$ → True

→ Here case 3 holds true.

Hence $T(n) = \theta(F(n))$ ⇒ $T(n) = \theta(n)$



Topic : Divide & Conquer

80%

71%

#Q. $T(n) = 9T(n/3) + n^{2.5}$

A $T(n) = \theta(n)$ ✗

B $T(n) = \theta(n\sqrt{n})$ ✗

C $T(n) = \theta(n \log n)$ ✗

D $T(n) = \theta(n^2\sqrt{n})$

Imp

$T(n) = \Theta(n^{2.5})$

$n^{2.5} = n^2 \times n^{0.5}$

$= n^2\sqrt{n}$



Topic : Divide & Conquer

$a = 9, b = 3, F(n) = n^{2.5}$ → Valid

$\log_b 9 = \log_3 9 = 2$

$n^2 * \sqrt{n}$

$n^2 * n^{0.5}$

$n^{(2 + 0.5)}$

$n^{2.5}$



Topic : Divide & Conquer

Case 1 : Is $n^{2.5} = O(n^{2-E})$, Some $E > 0$

$n^{2.5} \leq n^{2-E}$? $\rightarrow$ Not possible

$\rightarrow$ Fails

Case 2 : Is $n^{2.5} = \theta(n^2 * (\log n)^K)$, For some K

(a) $K \geq 0$ $\rightarrow$ Fail

(b) $K = -1$ $\rightarrow$ Fail

$n^{2.5} = n^2(\log n)^K$

$\sqrt{n} \neq (\log n)^K$

$\rightarrow$ Fails



Topic : Divide & Conquer

$f(n) = n^{2.5}$

$f(n/3) = (n/3)^{2.5}$

Case 3: Is $n^{2.5} = \Omega(n^{2+E})$ some $E > 0$

$n^{2.5} > n^{2+E}$, $E = 0.1, 0.2, 0.33, ...$

→ yes

and $a * F\left(\frac{n}{b}\right) \leq \delta * F(n)$ For some $\delta < 1$

$F(n) = n^{2.5}$ $a * F\left(\frac{n}{3}\right)^{2.5} \leq \delta * n^{2.5}$

$3^{2.5} = 3^2\sqrt{3}$

$\frac{9}{3^2\sqrt{3}} * n^{2.5} \leq \delta * n^{2.5}$

$\delta = 1/\sqrt{3}$

Case 3 is satisfied. $\Rightarrow$ $T(n) = \theta(F(n)) = \theta(n^{2.5})$



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Doubt : $F(n) = n^{2.5}$

$$F\left(\frac{n}{3}\right) \rightarrow \left(\frac{n}{3}\right)^{2.5}$$

$$n \rightarrow \frac{n}{3}$$

Validate all the D_nC recurrences that we have learnal so for using Master's method.

Back-sub



$$T(n) = 2T(n/2) + 2$$



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1. Min-Max Problem :

$T(n) = 2T(n/2) + 2$ → Value of Rec ⇒ $\left(\frac{3n}{2} - 2\right)$

TC → $O(n)$

→ $a = 2, b = 2, F(n) = 2$ (+ve) → Valid $\log_b a$

$\log_2 2 = 1$

Case 1 : Is $2 = O(n^{1-E})$, $E > 0$

Yes $E = 0.5, 0.1, 0.2, \ldots\ldots$ True

Hence $T(n) = \theta\left(n^{\log_b 9}\right) = \theta(n)$

B.S :- $T(n) = T(n/2) + c$



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2. Binary Search :

$T(n) = \theta(\log n)$

$T(n) = T(n/2) + C$

→ $a = 1, b = 2, F(n) = C$ → Valid $\log_b a = \log_2 1 = 0$

Check case 1

Is $F(n) = O\left(n^{\log_b a - E}\right)$ Some $E > 0$

$C = O(n^{0-E})$ Some $E > 0$ → Fails

~~$n - E, E > 0$~~ → Decreasing function



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Check case 2 :

Is $F(n) = \theta\left(n^{\log_b a} * (\log n)^K\right)$, some K

(a) $K \geq 0$

(b) $K = -1$

$C = \theta\left(n^0 * (\log n)^K\right)$?

If $K = 0$, $C = \theta(n^0 * (\log n)^0)$

$C = \theta(1) \rightarrow$ True

Hence case 2 (a) holds

$$T(n) = \theta\left(n^{\log_b q} * (\log n)^{K+1}\right)$$

$$= \theta(n^0 * (\log n)^{0+1})$$

$$T(n) = \theta(\log n)$$

$$T(n) = 2T(n/2) + O(n)$$



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3. Merge Sort : $TC = \theta (n \log n)$

$T(n) = 2T(n/2) + n$

Check $a = 2, b = 2, F(n) = n$ → Valid

$\log_b a = \log_2 2 = 1$

Check case 1 :

Is $F(n) = O(n^{\log_b a - E})$ some $E > 0$

$n = O(n^{1-E})$, some $E > 0$ → Fails

$n \leq n^{1-\epsilon}$



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Check case 2 :

Is $F(n) = \theta\left(n^{\log_b a} * (\log n)^K\right)$, Some K

→ (a) $K \geq 0$

(b) $K = -1$

$K = 0$, $n = \theta\left(n^1 * (\log n)^0\right)$

$n = \theta(n)$ → True

Case holds for $K = 0$

$$T(n) = \theta\left(n^{\log_b a} * (\log n)^{K+1}\right)$$

$$= \theta(n^1 * (\log n)^{0+1}) = \theta(n \log n)$$



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4. Matrix Multiplication :

(D_nC approach) $\rightarrow \theta(n^3)$

$T(n) = 8T(n/2) + n^2$

$a = 8,\ b = 2,\ F(n) = n^2,$ $\log_b a = \log_2 8 = 3$

Check case 1 :

Is $F(n) = O\left(n^{\log_b a - E}\right)$, some $E > 0$

Is $n^2 = O(n^{3-E})$, some $E > 0$?

$\rightarrow$ holds

$E = 0.1, 0.2, \ldots\ldots\ldots$ Yes.

Hence, $T(n) = \theta\left(n^{\log_b a}\right) = \theta\ (n^3)$



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5. Strassen's D_nC Matrix Multiplication :

$$T(n) = 7T(n/2) + n^2$$

$a = 7, b = 2, F(n) = n^2, \rightarrow$ Valid $\log_b a = \log_2 7 = 2.81$

Check case 1 :

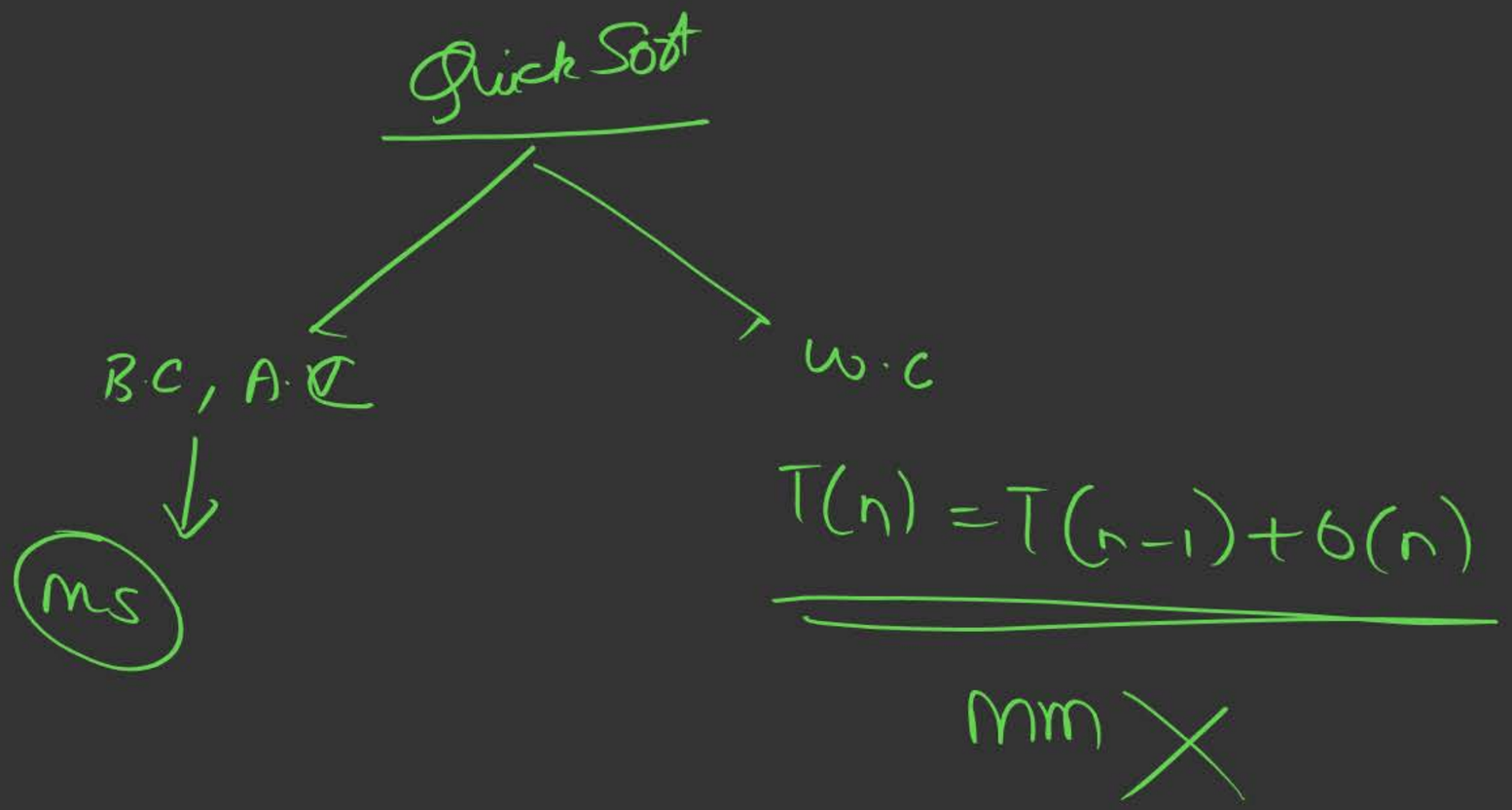
Is $F(n) = O\left(n^{\log_b a - E}\right)$, some $E > 0$

Is $n^2 = O(n^{2.81 - E})$, some $E > 0$?

$\rightarrow$ True

$E = 0.5, 0.4, \ldots\ldots\ldots$

Hence, $T(n) = \theta\left(n^{\log_b a}\right) = \theta\ (n^{2.81})$


Quick Sort
B.C, A.C
w.c
MS
T(n) = T(n-1) + θ(n)
mm



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Matrix mul

Master mtd



THANK - YOU